Types of Chemical Reactions

Answer key with worked solutions

A. Classifying

1. Classify.

> (a) **combination** (synthesis) (b) **decomposition** (c) **double displacement** (precipitation of $BaSO_4$) (d) **single displacement** (e) **combustion**

B. Predicting products

2. Will it react?

> (a) Mg is above Zn: $Mg + ZnCl_2 \longrightarrow MgCl_2 + Zn$
>
> (b) Ag is below Cu: **no reaction**
>
> (c) Zn is above H: $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$
>
> (d) Cl is more reactive than Br: $Cl_2 + 2\,NaBr \longrightarrow 2\,NaCl + Br_2$
>
> (e) Br is less reactive than Cl: **no reaction**

3. Complete and balance.

> (a) $2\,Na + Cl_2 \longrightarrow 2\,NaCl$ (b) $MgCO_3 \longrightarrow MgO + CO_2$ (c) $H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O$

C. Combustion

4. Combustion.

> (a) $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$
>
> (b) $2\,C_4H_{10} + 13\,O_2 \longrightarrow 8\,CO_2 + 10\,H_2O$
>
> (c) **Carbon monoxide** (CO), a poisonous gas (or soot, C).

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D. Double displacement

5. Why one reacts.

> Silver nitrate + sodium chloride gives **silver chloride, which is insoluble**, so a precipitate forms and the reaction happens: $\mathrm{AgNO_3 + NaCl \longrightarrow AgCl(s) + NaNO_3}$. With potassium nitrate + sodium chloride, the possible products (KCl and NaNO3) are **both soluble**: no precipitate, water or gas, so no reaction.

E. Calculations

6. Magnesium oxide.

> $$\begin{aligned} &2.43\ \cancel{\text{g Mg}} \times \frac{1\ \cancel{\text{mol Mg}}}{24.31\ \cancel{\text{g Mg}}} \\ &\quad \times \frac{2\ \cancel{\text{mol MgO}}}{2\ \cancel{\text{mol Mg}}} \times \frac{40.31\ \text{g MgO}}{1\ \cancel{\text{mol MgO}}} \\ &= \mathbf{4.03\ g\ MgO} \end{aligned}$$

7. Zinc used.

> $\mathrm{Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu}$: 1 mol Zn gives 1 mol Cu.
>
> $$\begin{aligned} &4.86\ \cancel{\text{g Cu}} \times \frac{1\ \cancel{\text{mol Cu}}}{63.55\ \cancel{\text{g Cu}}} \\ &\quad \times \frac{1\ \cancel{\text{mol Zn}}}{1\ \cancel{\text{mol Cu}}} \times \frac{65.38\ \text{g Zn}}{1\ \cancel{\text{mol Zn}}} \\ &= \mathbf{5.00\ g\ Zn} \end{aligned}$$

F. Challenge

8. Order of reactivity.

> **X > Z > Y**. A metal can only displace a less reactive metal. X displaces both others, so it is the most reactive; Z displaces Y, so Z is more reactive than Y.

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